

Machine Learning Based Disease Diagnosis & Medical Service System (ML-DDMSS)



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September, 2024

CERTIFICATION

It is certified that the following students have completed their project work on “**Machine Learning Based Disease Diagnosis & Medical Service System**”

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Session: 2020-2024

This report is submitted in partial fulfilment of the requirements of the award of degree of **Bachelor of Software Engineering**. Mirpur University of Science and Technology (MUST)

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ABSTRACT

In the evolving landscape of healthcare, integrating advanced technologies like Machine Learning (ML) and the Internet of Things (IoT) offers new opportunities to enhance medical services. This thesis presents the "Machine Learning-Based Disease Diagnosis and Medical Services System (ML-DDMSS)," a platform designed to transform healthcare interaction. By analysing real-time health data from smart-watch and user-reported symptoms, ML-DDMSS facilitates timely and accurate disease diagnosis. The system also uses Google Maps to recommend nearby hospitals specializing in diagnosed conditions, ensuring prompt medical attention. This research explores the feasibility of integrating ML with IoT, addresses data security, and evaluates regulatory compliance. The findings show that ML-DDMSS not only improves healthcare delivery but also holds the potential to significantly enhance patient outcomes, especially in remote or underserved areas.

UNDERTAKING

We certify that work titled “Machine Learning Disease Diagnosis & Medical Service System” is our own work. The work has not, in whole or in part, been presented elsewhere for assessment. Where material has been used from other sources it has been properly acknowledged referred.

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ACKNOWLEDGEMENTS

We would like to express our deepest appreciation to all those who provided me the possibility to complete this report. A special gratitude we give to our final year project Advisor, “Dr. Tehmina Shahryar”, whose contribution in stimulating suggestions and encouragement, helped us to coordinate our project especially in writing this report.

A special thanks to my team mates, who help to assemble the parts and gave suggestion about the Project “ML-DDMSS”. We have to appreciate the guidance given by another supervisor “Engr. Maryam Hamdani” as well as the panels especially in our project presentation that has improved our presentation skills thanks to their comment and advice.

“Team *SmartMed*”

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ABBREVIATIONS

Name	Abbreviation
<i>Software Requirement specification</i>	<i>SRS</i>
<i>Machine Learning-Based Disease Diagnosis & Medical Service System</i>	<i>ML-DDMSS</i>
<i>Internet of things</i>	<i>IOT</i>
<i>Sustainable Development Goals</i>	<i>SDGs</i>
<i>Artificial Intelligence</i>	<i>AI</i>
<i>Application Programming Interface</i>	<i>API</i>
<i>Final Year Project</i>	<i>FYP</i>
<i>Machine Learning</i>	<i>ML</i>
<i>Software Engineering Department</i>	<i>SED</i>
<i>Visual Studio Code</i>	<i>VS code</i>
<i>Database</i>	<i>DB</i>
<i>Microsoft</i>	<i>MS</i>
<i>Performance</i>	<i>P</i>
<i>Safety and security</i>	<i>S</i>
<i>Institute of Electrical and Electronics Engineers</i>	<i>IEEE</i>
<i>Unified Modelling Language</i>	<i>UML</i>
<i>Functional Requirement</i>	<i>F</i>

Chapter 1

INTRODUCTION

CHAPTER 1:

Introduction

In an era where technology is revolutionizing healthcare, our project, the "Machine Learning-Based Disease Diagnosis and Medical Services System (MLDDMSS)" aims to bridge the gap between patients, healthcare providers, and technology. Challenges such as overcrowded facilities, delayed diagnoses, and limited access to specialized care, especially in remote areas, have increased the demand for accessible and efficient healthcare services, highlighting the need for innovative solution.

We propose an innovative solution to offer convenient and timely medical services to individuals while alleviating the burden on healthcare facilities. This project combines the power of deep learning and the Internet of Things (IoT) to create a comprehensive application for disease diagnosis, health monitoring, and near-by hospital recommendations.

1.1 Goals

Followings are the goals of ML-DDMSS

- Develop an ML and IoT-driven healthcare platform that enables accurate disease diagnosis based on user-entered-symptoms and real-time health data collected from smartwatch.
- The platform will also help individuals by providing recommendations for nearby hospitals specialized in diagnosed disease, thereby facilitating timely and appropriate medical attention.
- In cases where smartwatches identify deviations in vital signs, users will receive notifications prompting them to input additional symptoms on app for a more precise disease diagnosis.

This collective approach aims to bridge the gap between users and healthcare resources, ultimately improving overall healthcare accessibility and outcomes. It's not only about serving urban populations also rural areas, making a difference where it's needed most.

1.2 Motivation

The goal of the Machine Learning-Based Disease Diagnosis and Medical Services System (ML-DDMSS) is to increase healthcare efficiency and accessibility, particularly

in places where it is frequently deficient. Many face obstacles such as delayed diagnosis, overcrowding in hospitals, and trouble getting specialist care. There's a great chance to employ technology to close these gaps as smartwatch adoption grows. We developed ML-DDMSS since there was no platform that combined real-time health monitoring, precise disease diagnosis, and prompt hospital advice. In order to relieve burden on healthcare systems and improve patient care, we want to provide users with instant health insights and direct them to the appropriate medical facilities.

1.3 Method

The ML-DDMSS project employs a modular construction methodology to effectively handle the system's extensive functionality and intricate design. The work has been divided into several major elements, such as integration with smartwatches, prediction of local hospitals, and diagnosis. This method provides flexibility in both development and testing by allowing us to focus on each component separately. We utilize machine learning models, which are well-known for their capacity to automatically extract characteristics and produce accurate results, to guarantee accuracy and manage large-scale data efficiently. To create a cross-platform app that runs on both iOS and Android and is usable by a wide range of users, we have opted to use the Flutter framework.

With three people on our team, we can work together effectively because we divide up the workload. One module at a time, we can concentrate on before combining all of the components into the entire system, make sure they have all been fully satisfied. This approach fits the objectives of the project and our team structure.

1.4 Problem Statement

Even in today's era of technology, many people face delays in diagnosis of their disease, overcrowded hospitals situations, and limited access to specialized healthcare, especially in remote areas. Many patients suffer long-term consequences or fail to recover fully just due to late diagnosis of disease. Additionally, there is a lack of integrated systems that combine smartwatch data with symptoms-based diagnosis. Furthermore, patients also struggle to find suitable nearby hospitals specializing in their diagnosed conditions, often lacking the knowledge to locate the best available care.

1.5 Report Overview

The report introduces a system that allows the reader to study the system's basic elements. The report contains five chapters.

Chapter 1

It reviews the system's introduction and basic motivations. It is an overview of our system, including the problem statement and methodology used to develop a project. This chapter covers the main notions and system abstractions.

Chapter 2

It provides an overview of the project's background. It provides an overview of the existing similar systems and published research papers in this category till yet. This section will review literature to better understand and explain essential success criteria and feasibility. Additionally, it highlights how our system is more good than existing systems.

Chapter 3

This chapter is about software requirement specification, highlight the basic functional and non-functional requirements need to develop a system or to solve the problems in existing systems as reviewed in previous chapter. It provides basic overview of system architecture and use cases of our systems.

Chapter 4

This chapter is about system design and architecture, highlight the system's architectural diagrams like Class Diagram, Use case Diagram, Sequence Diagram, DFD, State Machine and Deployment Diagram.

Chapter 5

This chapter reviews the software design. It provides an overview of user interfaces (UI) and interaction of user with our product.

Chapter 6

This chapter reviews the implementation of system. It provides an overview of development environment, dataset used, machine learning model, languages and technologies used to develop a system. Additionally, it also highlights the challenges faced during development and how we handle them.

Chapter 7

This chapter reviews the testing and validation of our product. It highlights the types of testing done i.e ing, integration testing and system testing. Firstly, system is validated by the development team using a manual testing to check against requirement specification document.

Chapter 8

This chapter closes the project by providing the overview about the conclusion and findings. It also highlights the limitations in our project and future work in this category.

Chapter 2

LITERATURE REVIEW

CHAPTER 2:

Literature Review

2.1 Introduction

The literature review explores the current state of research and working in the field of health care and disease diagnosis. In this chapter we identify the limitations of existing systems and the need for more integrated and comprehensive solutions that combine real-time health data collection with symptoms based diagnosis and nearby hospital recommendation.

Various digital health platforms have emerged, offering symptom based diagnosis or sensor based health monitoring. These platforms use artificial intelligence (AI) to analyse user data, yet they often lack the integration of smart-watch for continuous real-time health monitoring, which is more convenient to use than wearing separate sensors. Our proposed system, ML-DDMSS, distinguishes itself by integrating machine learning models with IoT devices like smartwatches, continuously monitoring health metrics and offering real-time diagnosis. Additionally, ML-DDMSS enhances user experience by recommending nearby specialized hospitals using Google Maps, a feature that further sets it apart from existing solutions.

2.2 Literature Review

Several AI-driven health platforms have been developed, each with its unique approach to improving healthcare delivery:

- **Ada Health:** Ada Health employs AI within a mobile app to assess user symptoms and provide personalized health information. It relies on a comprehensive database and algorithms to analyse user-entered symptoms.
- **Google Health's DeepMind:** DeepMind by Google Health uses deep learning techniques to predict patient outcomes, analyzing relevant medical data through advanced algorithms.

- **Buoy Health:** Buoy Health offers a Chabot-based platform that uses AI to assess symptoms in a conversational manner, providing potential explanations for users' health concerns.
- **IBM Watson for Oncology:** IBM Watson for Oncology supports oncologists in diagnosing and treating cancer. It analysis vast amounts of medical data using AI to recognize patterns in various types of cancer.

Despite the advancements these platforms represent, they primarily focus on either symptom analysis or specific areas like oncology. Offer continuous health monitoring through different sensors but none offer the continuous health monitoring using smart-watch, and integrated hospital recommendations. All the above gaps are filled by ML-DDMSS, combining the capabilities of smart-watch and deep learning to address the full spectrum of healthcare needs—from monitoring and diagnosis to finding the most suitable nearby hospital for that disease.

In contrast to these existing systems, ML-DDMSS provides a more comprehensive and user-friendly solution. The platform's integration of smart-watch for continuous health monitoring, real-time disease diagnosis, and nearby hospital recommendations makes it more valuable particularly for remote populations than existing systems.

Following is the comparison table of existing health systems:

Table 2.1 Literature Review Existing Health Systems

Parameter	Ada Health	Google Health's DeepMind	Buoy Health	IBM Watson for Oncology
Date of Publication	2016	2016	2012	2013
Working	Symptom checker, personalized health insights	Medical research, drug discovery, image analysis	Symptom checker, triage support, personalized care plans	Cancer diagnosis and treatment support

Technologies Used	Natural language processing, machine learning, artificial intelligence	Deep learning, machine learning, natural language processing	Natural language processing, machine learning, clinical decision support systems	Machine learning, natural language processing, data mining
Diseases Tackled	Wide range of common and rare conditions	Focus on complex diseases like cancer and neurological disorders	Common ailments, chronic conditions, mental health	Wide range of cancers
IoT Devices Used	No	Can integrate with medical devices and wearables	No	Can integrate with medical devices and wearables
Hospital Suggestion	No	No	Can suggest specialists and healthcare providers	Can suggest clinical trials and treatment options

Following are the recent journals and conferences related to ML-DDMSS:

Table 2.2 Literature Review Research Papers

Title	Year	Dataset	Methodologies/ Algorithms	Evaluating Measure	Limitations	Product	Number of predicted Disease
MedAi: A Smartwatch-Based Application Framework for the Prediction of Common Diseases Using Machine Learning [1]	2022	properity (250 enteries)	Random forest, Support vector machine & Knearest neighbors	99.1%, 89.81% and 99.33% respectively	1. Small dataset. 2. Limited diseases prediction. 3. Only based on the collected data by smartwatch	Application+Wearable(Osense watch)	12
Disease Prediction using Symptoms based on Machine Learning Algorithms [2]	2022	Kaggle: (4921 instances)	Decision trees, Naive Bayes, and Random forest classifiers.	94%,93% and 98%	1. complex Interface with 6 dropdowns with 133 options each.(2). Small dataset. (3). No Realtime data collection	One screen UI build with flask	41
Symptom-based Disease Prediction Using Machine Learning [3]	2024	Kaggle: Disease Prediction Using Machine Learning (4920 instances)	Multinomial Naive Bayes, Decision Tree, and Random Forest	100%, 100% and 97.62% respectively	1. Small dataset. 2. Limited diseases prediction. 3. Only give disease based on the information entered by the user. No wearable included	Mobile application	42

A Precision Health Service for Chronic Diseases: Development and Cohort Study Using Wearable Device, Machine Learning, and Deep Learning [4]	2022	properity (1,667 patients data)	Decision trees, random forests, linear discriminant analysis, AdaBoost and adaptive boosting	78.2%,91.4%,82.6%,88.2% and 92.3%	1. Predict only chronic disease 2. Not completely automated, include doctors to analyse data and predict disease. 3. Using sensors for environment information. What if the environment information is not available. (limited usage)	Mobile application, Wearables and IOT in home for environment information	Chronical diseases
Simulation of Wearable Smart Jewelry Health Monitoring System Based on Decision Tree Algorithm [5]	2023	Properity (200 entries)	DT(Decision tree) algorithm	95.45%	Complexity in connecting multiple devices through mobile phones	Wearable smart jewelry (smart brooch)	30
Me, My Health, and My Watch: How Children with ADHD Understand Smartwatch Health Data [6]	2023	Not Specified	Smartwatch Visualizations, Personal Informatics Systems	Need for accessible visualization for children with ADHD, Importance of engagement in personal informatics research	Personal informatics systems for children with ADHD	Not specified	Not specified
Mobile Application Development for Disease Diagnosis based on Symptoms using Machine Learning Techniques [7]	2023	Kaggle: Disease Symptom Prediction (6000 instances)	K Nearest Neighbours, Support Vector Machine, Decision Tree, Random Forest Classifier, Naive Bayes and Artificial Neural Network	98.84% accuracy	No real time health monitoring (no wearable), limited diseases prediction	Mobile Application	40
Implementation of Efficient Machine Learning Techniques for Prediction of Cardiac Disease using SMOTE [8]	2024	UCI Machine Learning Repository (303 instances)	Random Forest (RF), Naive Bayes (NB), AdaBoost (AB), Decision Tree (DT), Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and Logistic Regression (LR)	96.6% accuracy	1. Small dataset. 2. Limited diseases prediction. 3. No wearable included	No, only train and test data using seven ML algo	1
An Intelligent Recommender System using Machine Learning Association Rules and Rough Set for Disease Prediction from Incomplete Symptom Set [9]	2024	Kaggle: Autism dataset(4920 rows)	Six popular ML algorithms such as AR, Decision Tree (DT), Random Forest (RF), KNearest Neighbors (KNN), Linear Support Vector Machine (LSVM), and Naive Bayes	97.9% accuracy	1. Small dataset. 2. Limited diseases prediction. 3. Only give disease based on symptoms. No wearable included	No, only train and test data using six ML algo	41
IoT-Based Smart Health Monitoring System for COVID19 [10]	2022	Not mentioned	No specific Machine Learning algo is mentioned	1.05% (98.95% accuracy)	Limited data collection, Less security(data spoofng could be done), Data transmission without encryption. Only predict Covid	App & Web application + Health Sensors	1
Intelligent health management based on analysis of big data collected by wearable smart watch [11]	2023	Not mentioned	No specific Machine Learning algorithm is mentioned	health, subhealth, or disease	No authentic proof of TCM, needed further scientific evidencebased study	Mobile Application + ANDUN health watch and Sensors	Not mentioned

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

CHAPTER 3:

Software Requirement Specification (SRS)

3.1 Introduction

The Software Requirements Specification (SRS) chapter provides a comprehensive overview of the system's requirements and specifications. This chapter lays the foundation for the development of the ML-DDMSS by detailing the necessary functionalities, performance metrics, and constraints. It identifies the core requirements that the system must fulfil, both from a functional and non-functional perspective, ensuring that the development process is guided by a clear and precise blueprint. This chapter also highlights the system architecture and use cases of MLDDMSS. The SRS chapter serves as a critical reference point for the development team, helping to align the project's goals with its technical implementation.

3.2 Functional Requirements

Following are the core functions of ML-DDMSS:

- **F1:** The system shall provide user registration functionality, allowing users to create securely.
- **F2:** The system will have a sign-in option for both users and admins to log in using their available username along with password or they can unlock through their Google account also.
- **F3:** The system shall allow users and admins to change or restore the password after email or phone number OTP verification.
- **F4:** Users will be able to enter their symptoms into a smartphone app, while the system analyses their health data on a connected Smartwatch.
- **F5:** The system shall use a machine learning model to analyse symptoms and display a predicted disease to the user on app or website.
- **F6:** The system shall allow users to download their medical record in pdf format.

- **F7:** The system shall allow users and admins to delete their accounts from the system.
- **F8:** The system shall allow users and admins to log out their accounts from the system.
- **F9:** The system shall recommend nearby best hospitals and clinics based on predicted health conditions using Google Map.
- **F10:** All notifications will be sent directly to the user's smartwatch, updating them on symptom entries and disease predictions as well as treatment recommendations. In cases of critical medical conditions, the system will send out immediate emergency alarms aiding with rapid intervention.
- **F11:** The system shall integrate seamlessly with Samsung Smart-Watch for real-time data collection and notify users in case of any irregular health metrics.
- **F12:** The system shall allow admins to view the number of users registering on the system.
- **F13:** The system shall allow admins to deactivate or delete the user's account.
- **F14:** The system shall allow admins to analyse the daily activities on the system.

3.3 Non-Functional Requirements

Following are the non-functional requirements for the ML-DDMSS, as specified using fit criteria:

Performance Requirements

- **P1:** The system must process symptom inputs and health data for disease prediction within 1-2 minutes.
- **P2:** The system shall monitor and collect health data Samsung Smart-Watch continuously without delay.
- **P3:** The system shall notify users of any irregular health metrics, such as a rise in blood pressure, in 10 sec upon detection.

- **P4:** The system shall recommend nearby hospitals and clinics based on predicted health conditions within 1 minute of the request.
- **P5:** The system shall predict disease within 1 minute after input.
- **P6:** The system shall authenticate user access securely within 5 seconds.
- **P7:** The system shall allow users to login and logout within 5 seconds.

Safety and Security Requirements

- **S1:** Secure authentication-A system will be in place to ensure that data is secure and not accessible by unauthorized personnel both at city end as well user side.
- **S2:** User health data from IoT devices will be kept private through strong encryption during transmission and storage.
- **S3:** The system will comply with data protection regulations like GDPR to safeguard user data.
- **S4:** Safeguards will be in place to prevent unauthorized access and ensure data integrity, following the principle of least privilege
- **S5:** The system shall provide user identity authentication to ensure that only authorized users can access the system and its features.
- **S6:** The system shall implement secure communication protocols for data transmission between the mobile application and IoT devices to prevent data breaches.

Software Quality Attributes

- **Availability:**
- The system shall be available 24/7 when needed, ensuring minimal downtime of 5 days per year.
- Automated backup and recovery processes shall be implemented to ensure data integrity and availability in case of system failure.
- **Usability:**

- The system will have an intuitive, user-friendly interface following best practices in usability from the industry.
- **Performance/Response Time:**
- Normal 8 campus tasks will respond to the user within less than 10 seconds and there is no more complex processing is taking place over two minutes.
- **Maintainability:**
- The system will be industry best practice and built to follow standards, easy maintenance and updates include developer's documentation.
- **Compatibility:**
- The system will be compatible with major web browsers (Chrome, Firefox, Safari, Edge) to ensure a consistent user experience across platforms.
- The system will support common operating systems (Windows, iOS, Android) to enable access from a variety of devices.

3.4 Use Cases

Key use cases include:

- **User Registration (U1):** You will be registered with the system and then you can.
- **Login (U2):** Allows users login into the system. Extended by login with Google and password restored in case of forgetting.
- **Input Symptoms (U3):** Enables users to input their symptoms on basis of Chatbot questions.
- **Receive Disease Prediction (U4):** Provides users a possible disease based on their symptoms.
- **Download History (U5):** Allows users to download their medical history.
- **Logout (U6):** Secures the user's session and ends it upon request.

- **View and Track the Recommended Hospitals (U7):** Show the relevant hospital in Google maps so users can also see how close it is.
- **Delete Account (U8):** Have access to users and admins delete accounts.
- **Collect Health Data (U9):** Records and saves health-related information from smartwatches.
- **Deactivate Account (U10):** User Account Deactivation for Admins.
- **Analyse Daily Activities (U11):** Allows system administrators to monitor and analyse user activity.
- **View Users (U12):** Provides system administrators with a view of all users in the system.

3.5 System Architecture

The system architecture describes systematically, the Component with in broader picture of how they are interconnected. It is designed to support modular development, which allows each component to be built individually. The architecture includes the following components:

- **Data Collection:** Handles the collection of health data from smartwatches and user inputs.
- **Disease Diagnosis:** Processes the collected data to generate disease predictions.
- **Recommendation:** Provides users with hospital recommendations based on their predicted condition and location.
- **User Interface:** Manages the interactions between the user and the system, including symptom input, data visualization, and receiving notifications.

Chapter 4

SYSTEM DESIGN

CHAPTER 4:

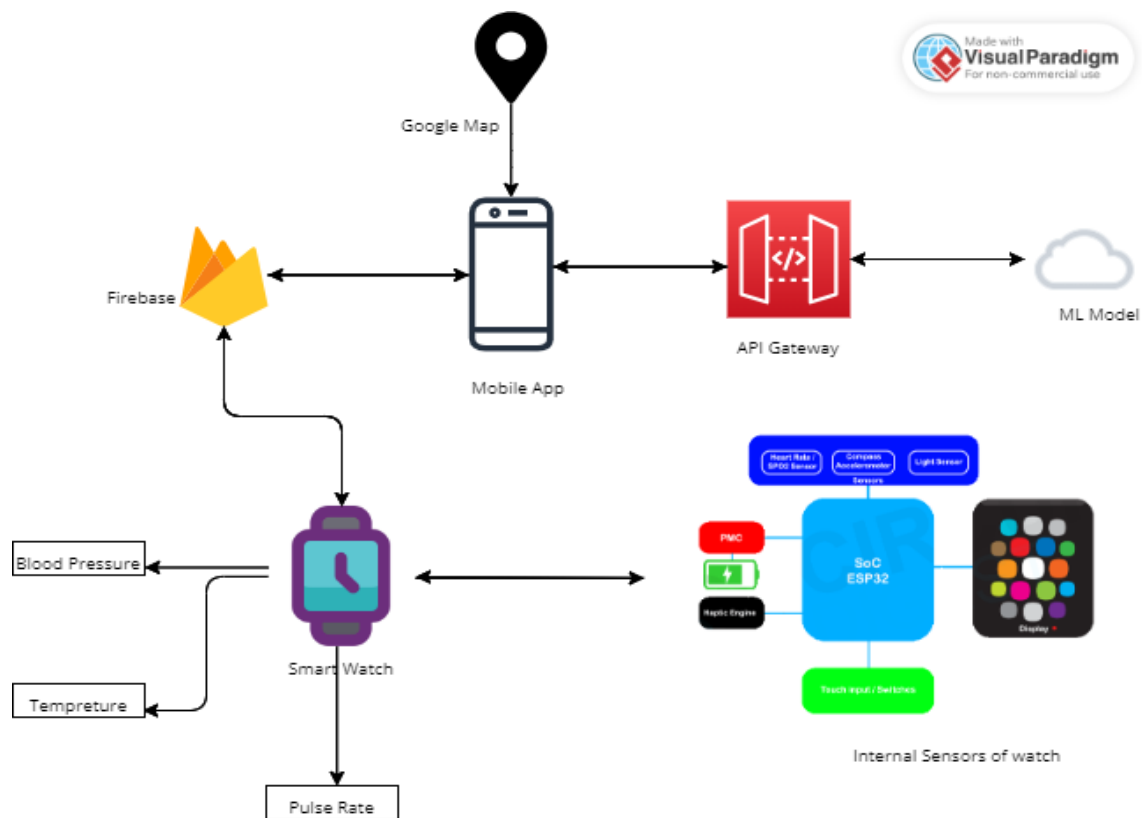
Design

4.1 Introduction

This chapter outlines the detailed information about the high level and low level design of a system. It includes diagrams, models that demonstrate the flow of information and interaction among different components of system.

4.2 System Design

The system design focus on the overall design of ML-DDMSS system, represents main components and their interactions. It shows the generic flow of information and working of a whole system.



4.3 Architectural Diagrams

This section includes diagrams that demonstrate the system architecture and design, such as:

Class Diagram

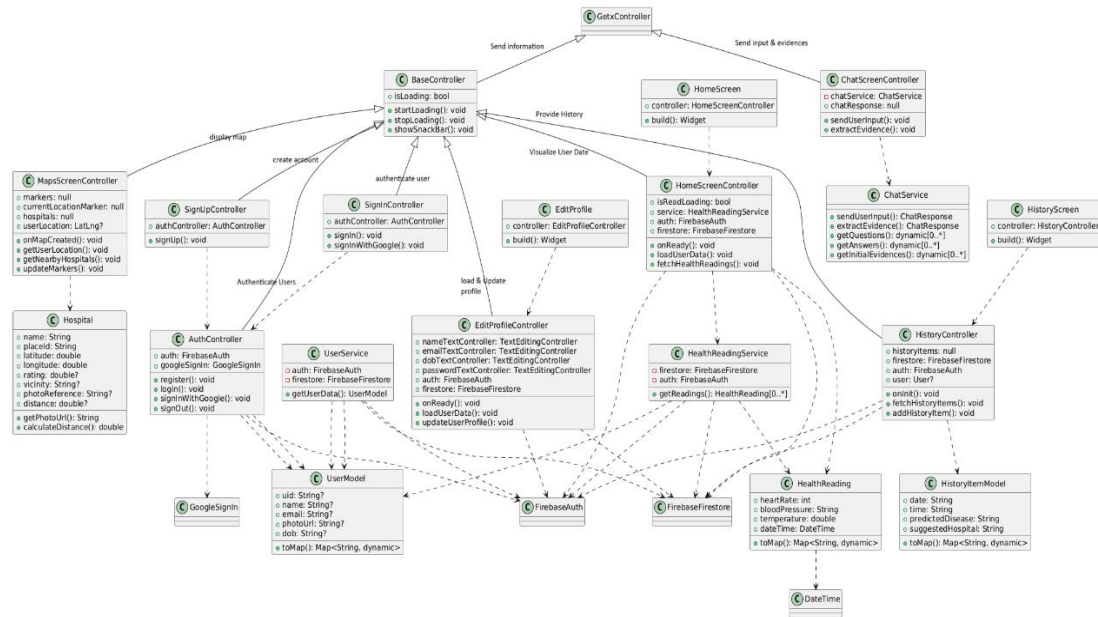


Figure 4.2 Class Diagram

State diagram

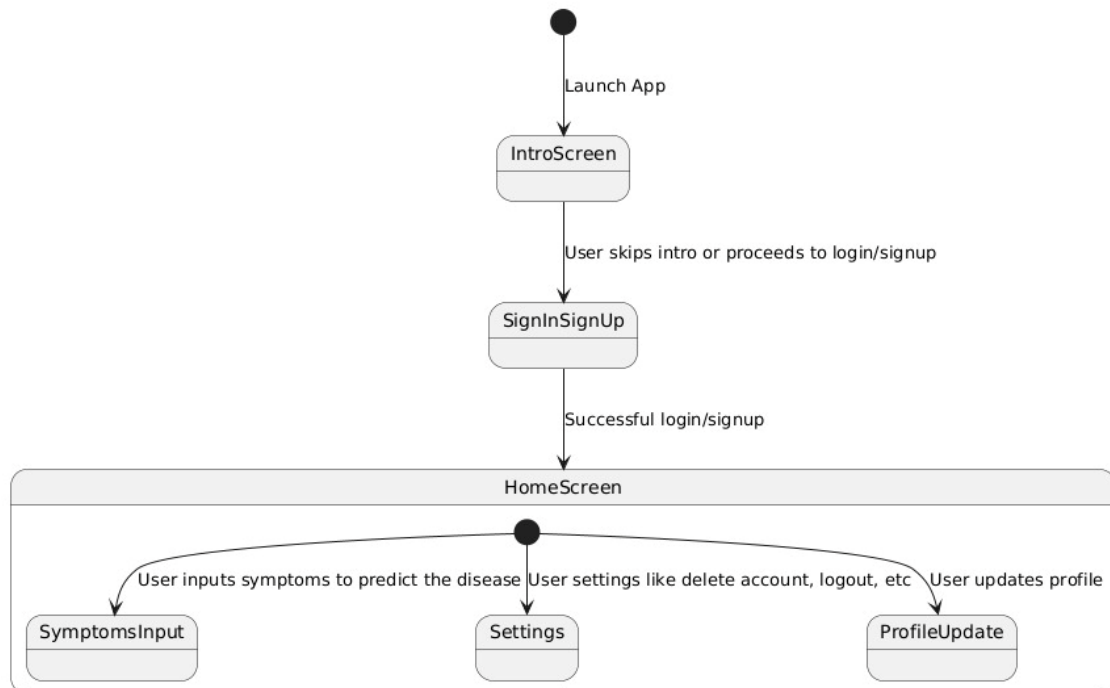


Figure 4.3 State Diagram

Use case

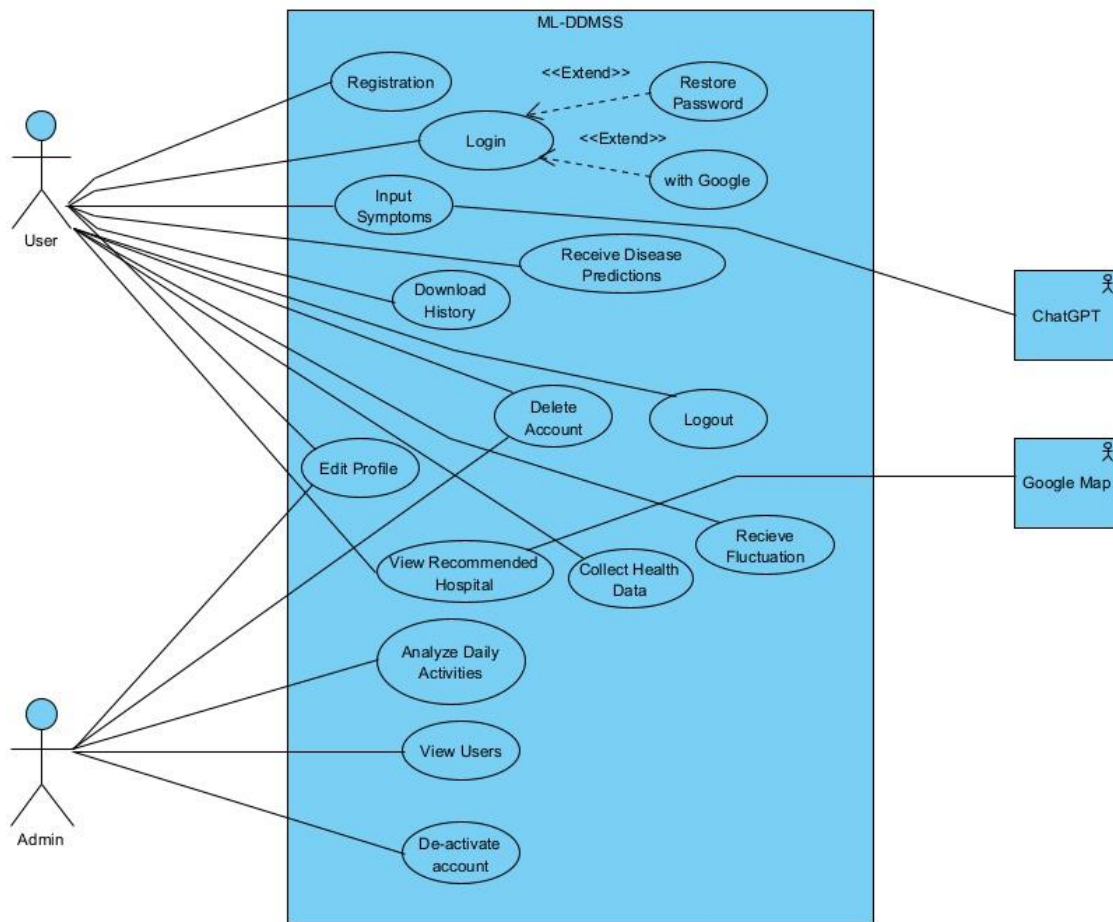


Figure 4.4 Use Case Diagram

DFD

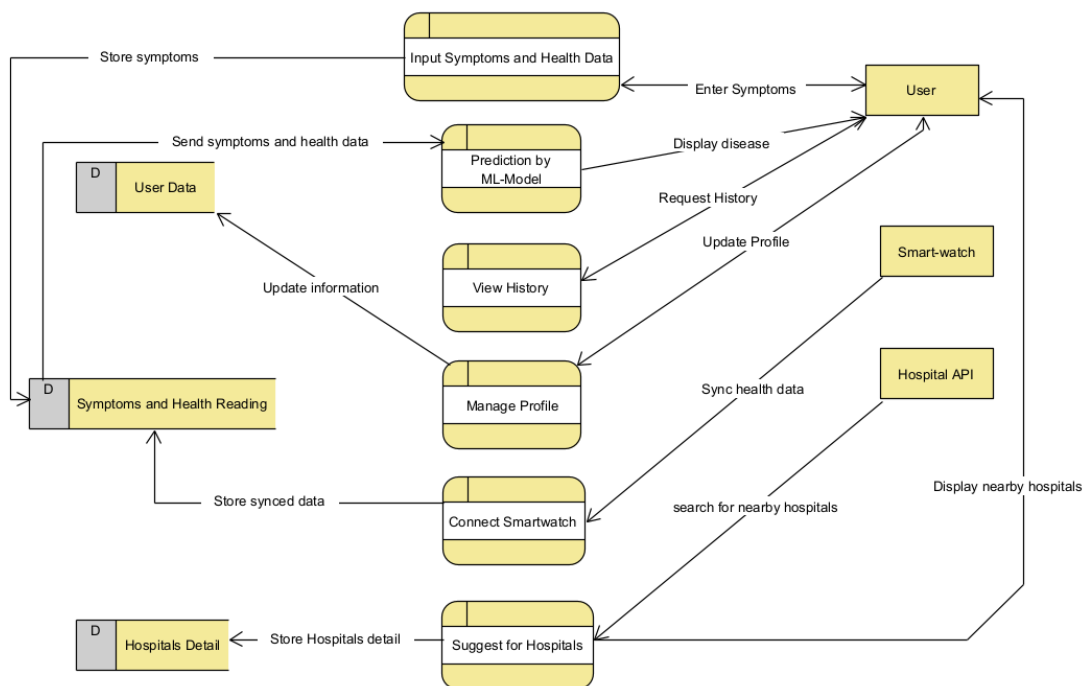


Figure 4.5 DFD Diagram

Sequence Diagrams

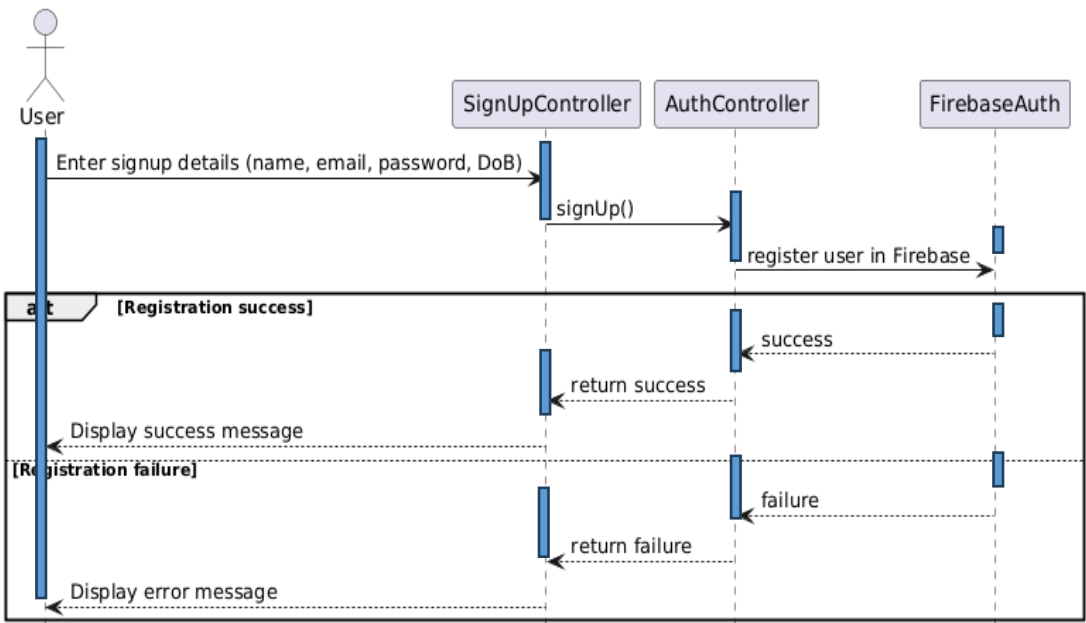


Figure 4.6 Sequence Diagram Sign up

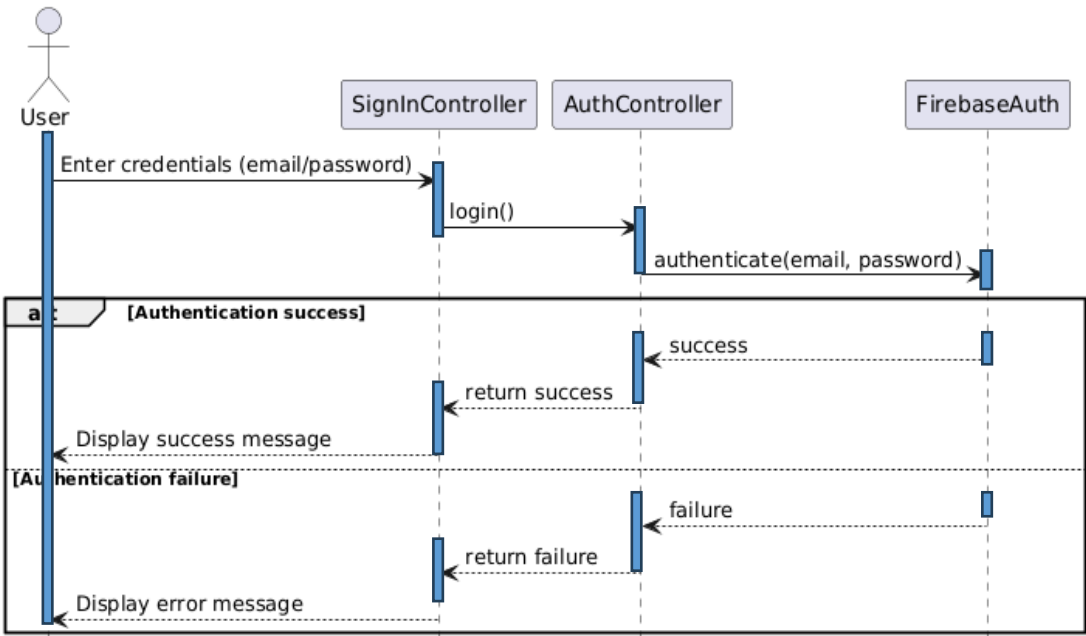


Figure 4.7 Sequence Diagram Login with Email

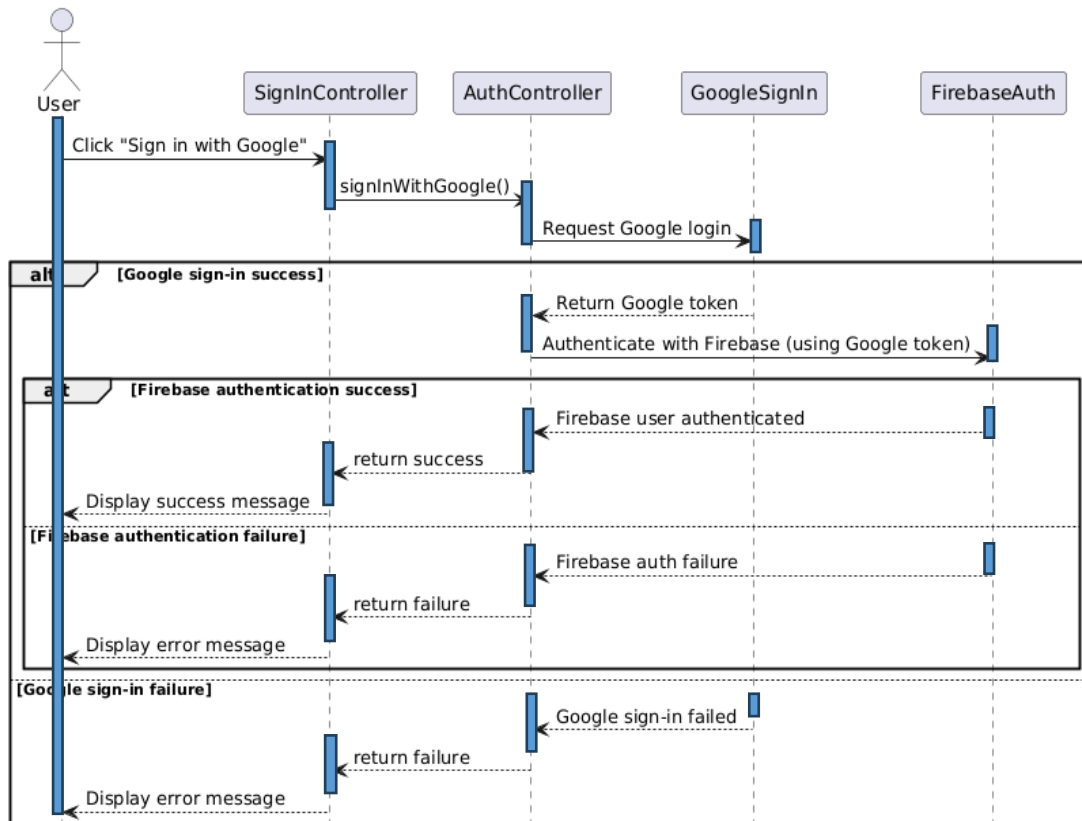


Figure 4.8 Sequence Diagram Login with Google

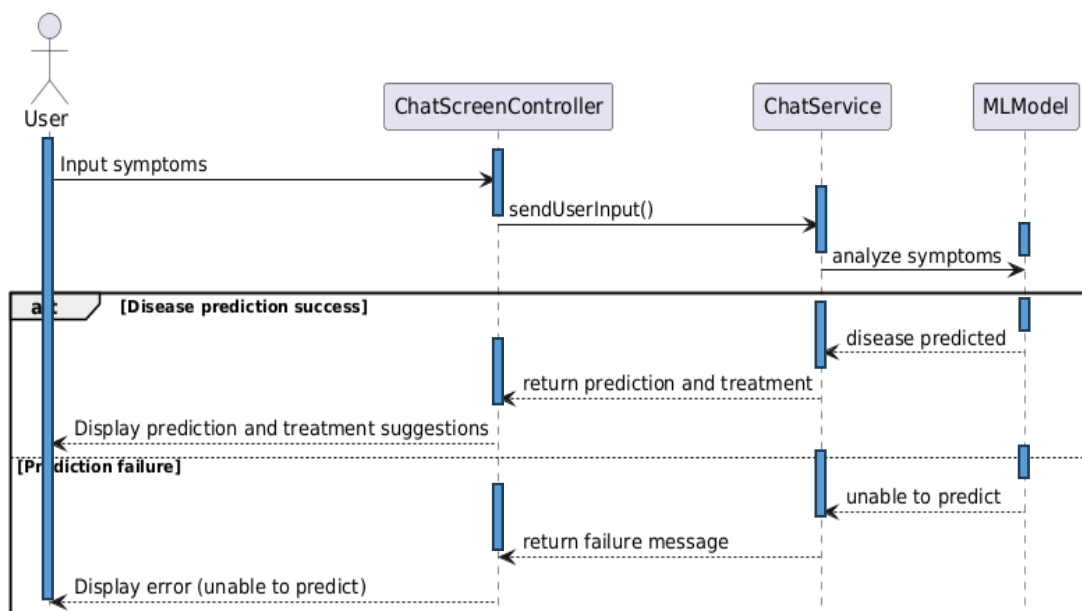


Figure 4.9 Sequence Diagram Input Symptoms

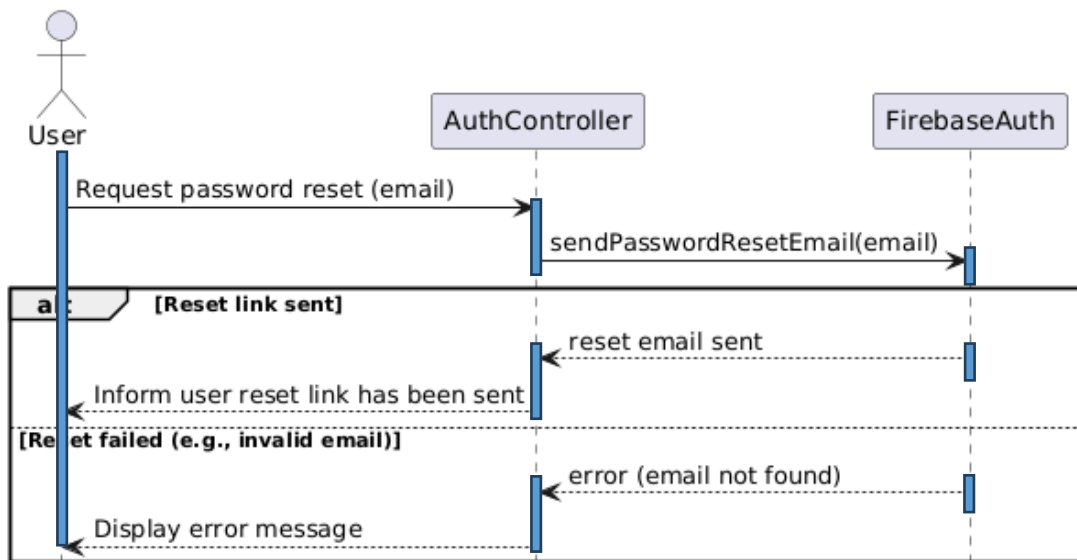


Figure 4.10 Sequence Diagram Reset Password

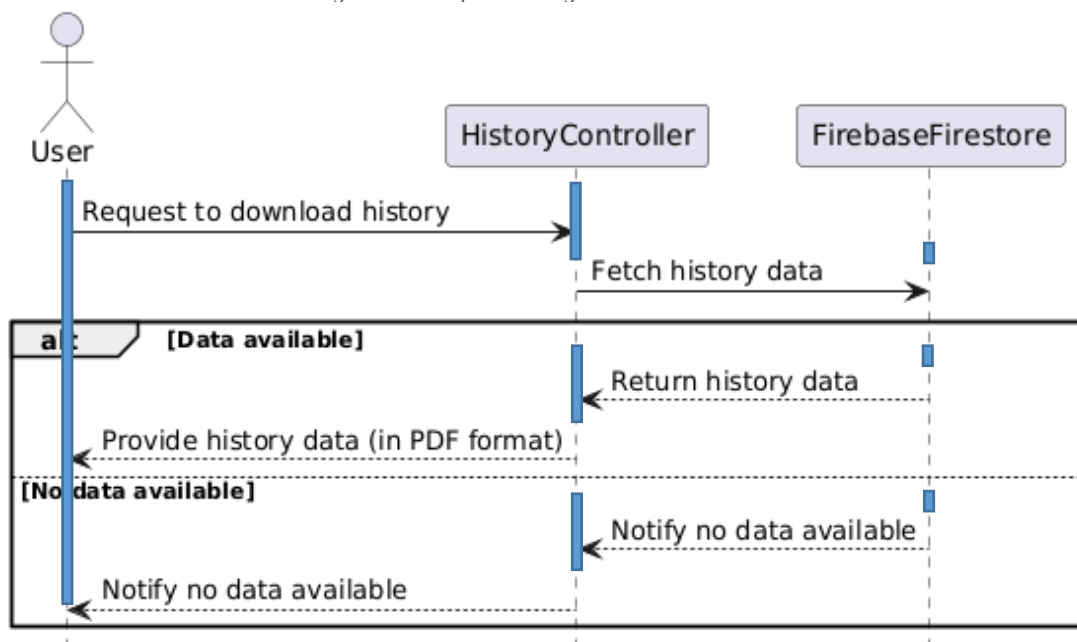


Figure 4.11 Sequence Diagram Download History

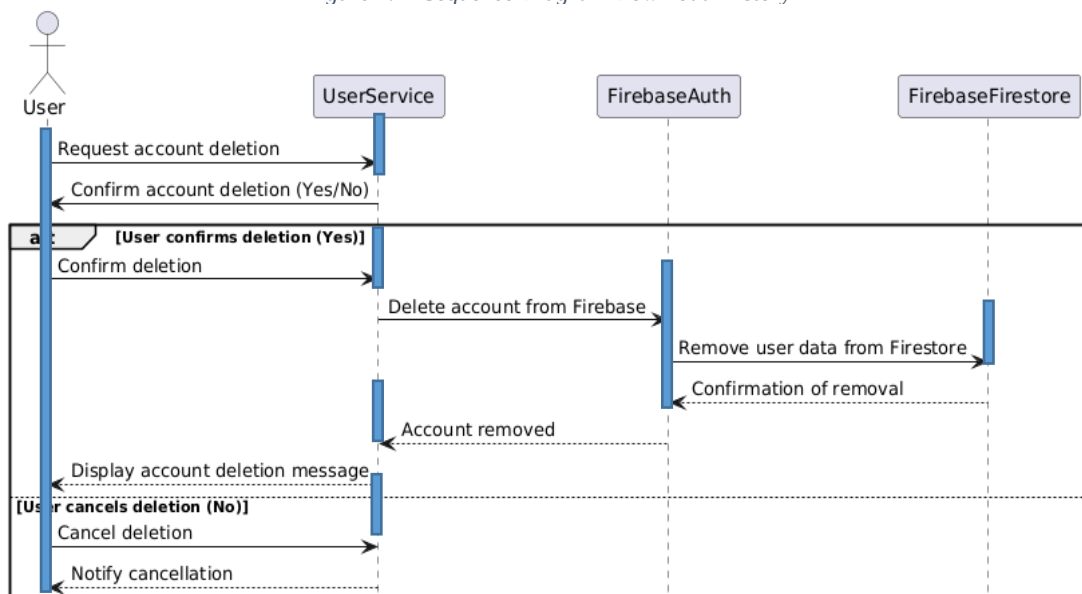


Figure 4.12 Sequence Diagram Delete account by user

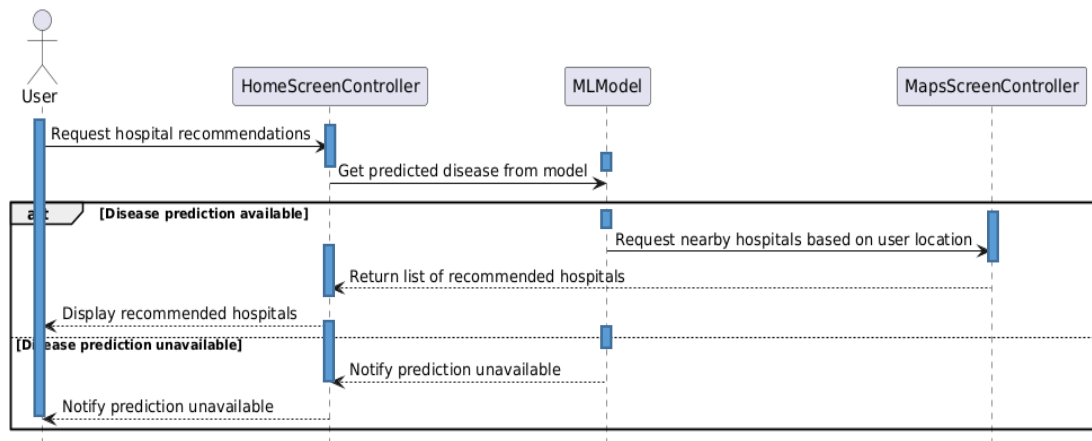


Figure 4.13 Sequence Diagram Hospital Recommendation

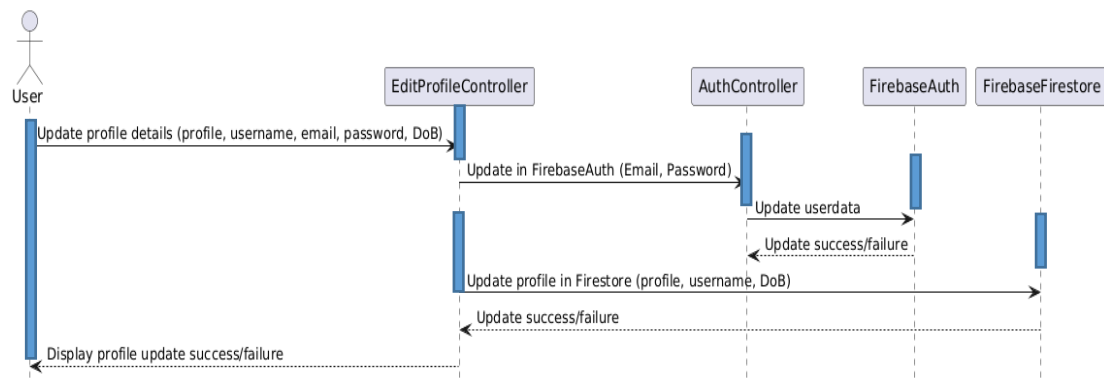


Figure 4.14 Sequence Diagram Update Profile

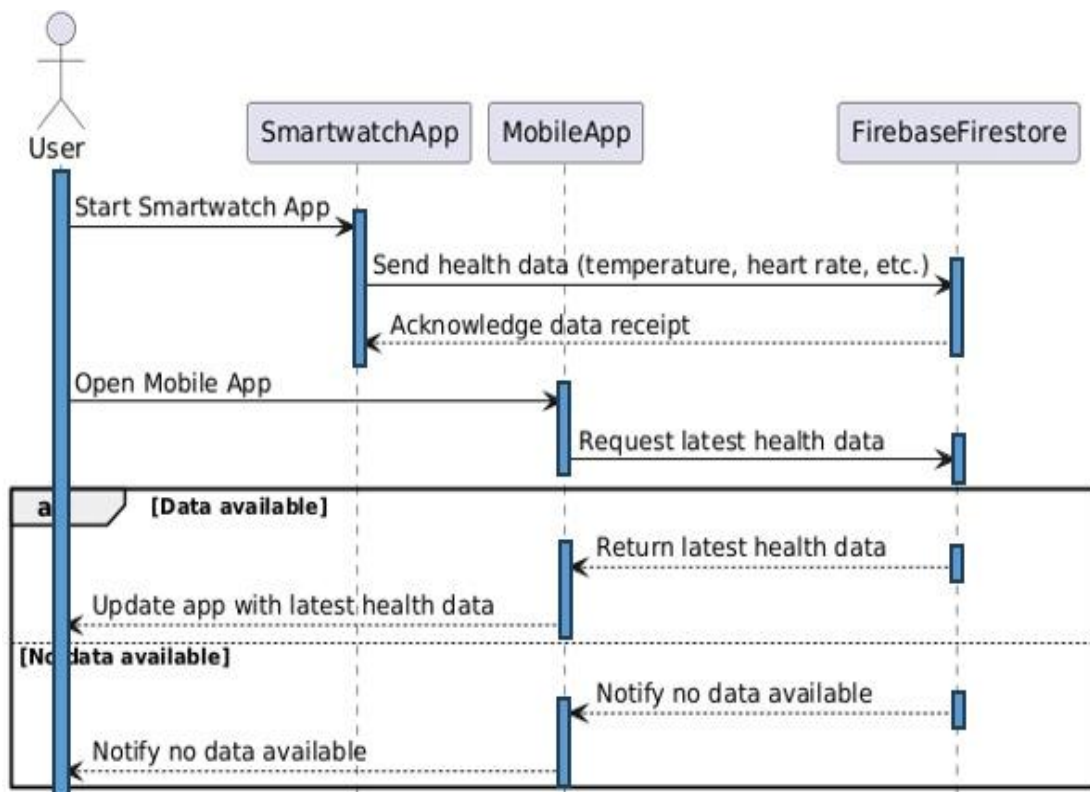


Figure 4.15 Sequence Diagram Collect Health Readings

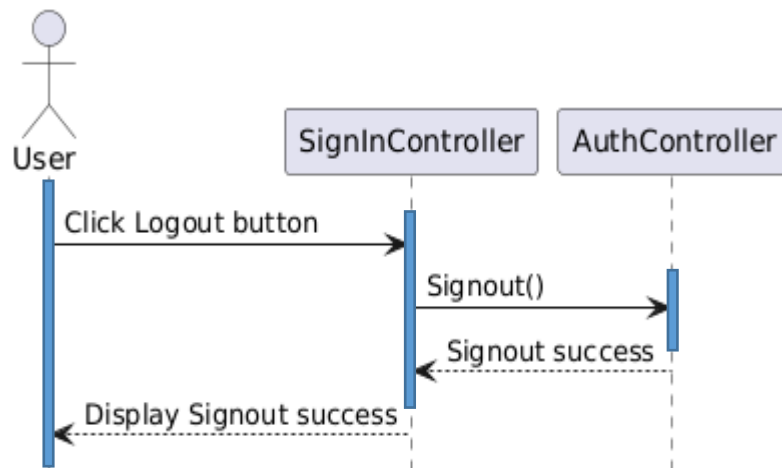


Figure 4.16 Sequence Diagram Logout

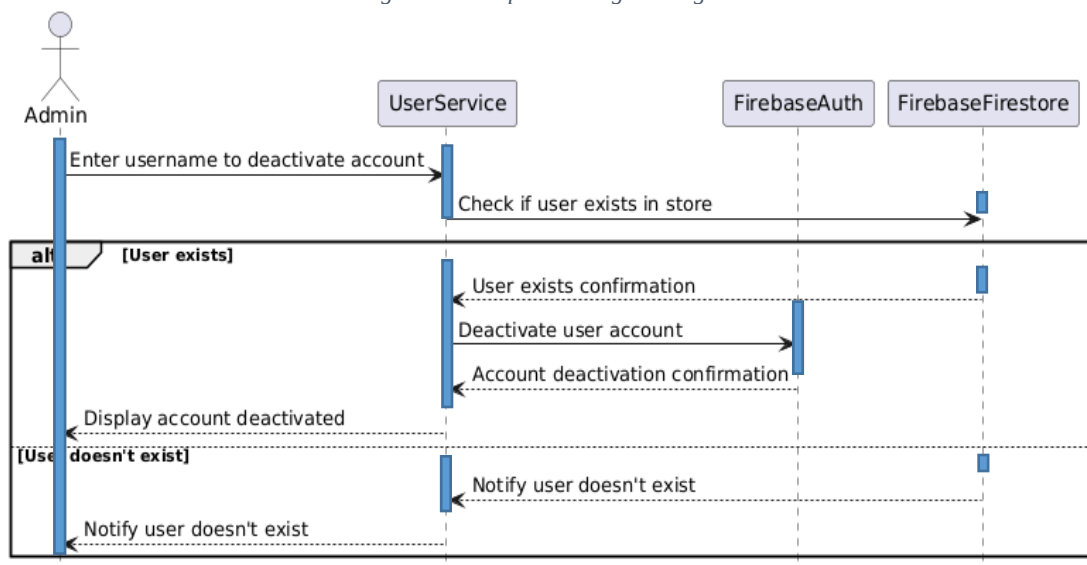


Figure 4.17 Sequence Diagram Deactivate user account

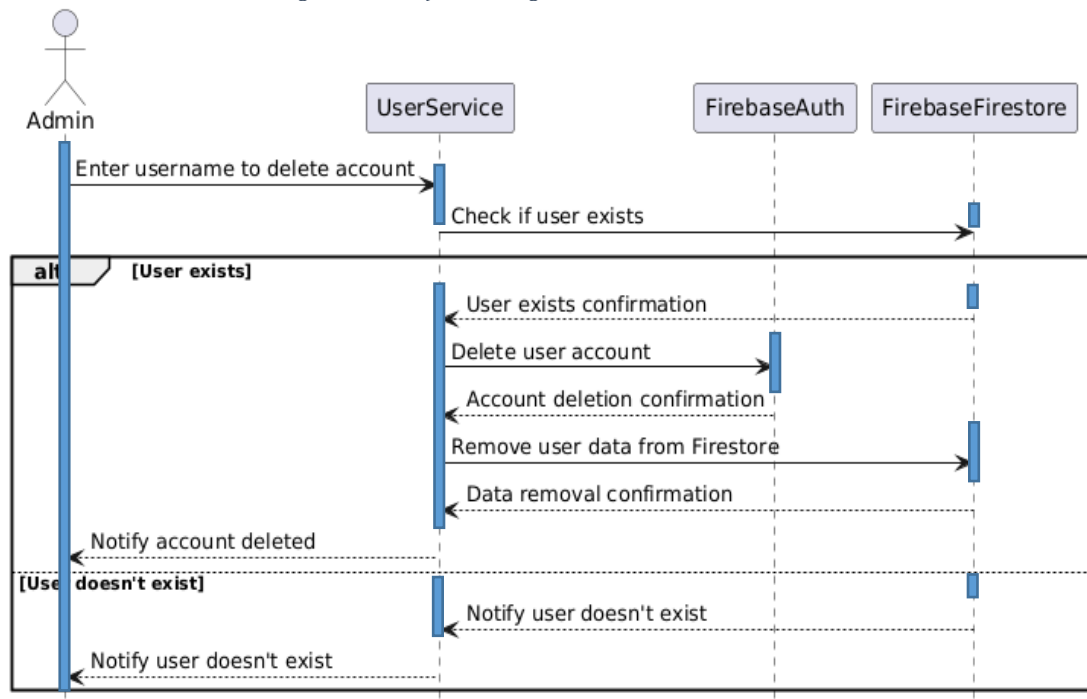


Figure 4.18 Sequence Diagram Delete user account by admin

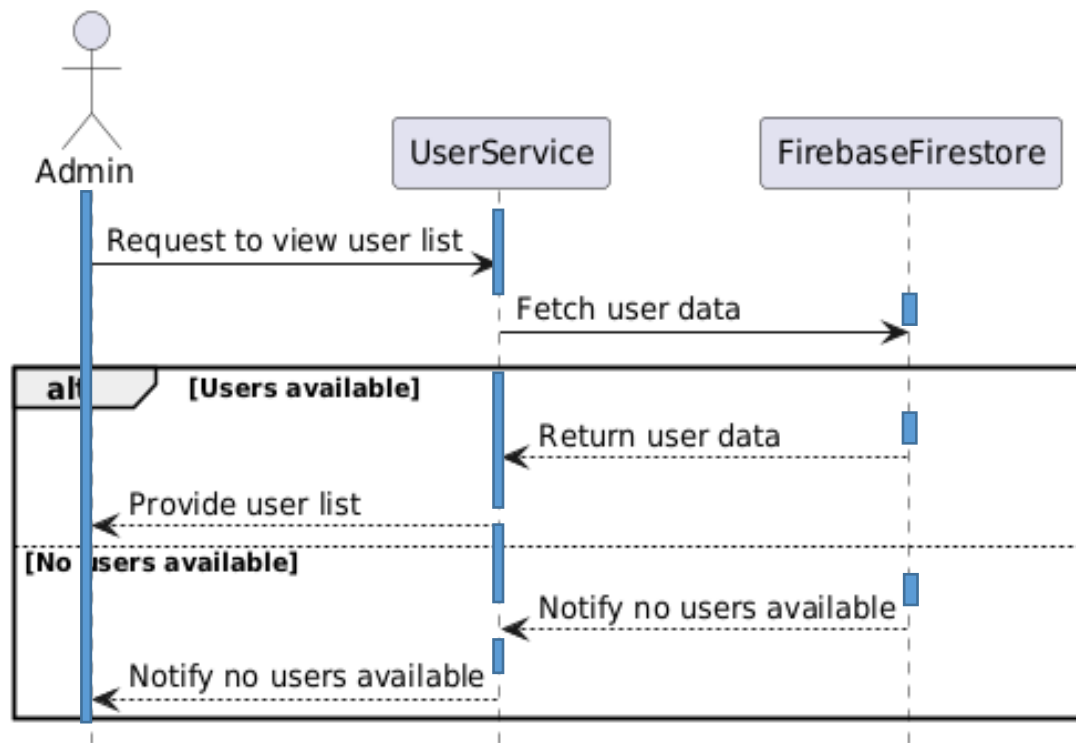


Figure 4.19 Sequence Diagram View Users

Deployment (for hardware-based projects)

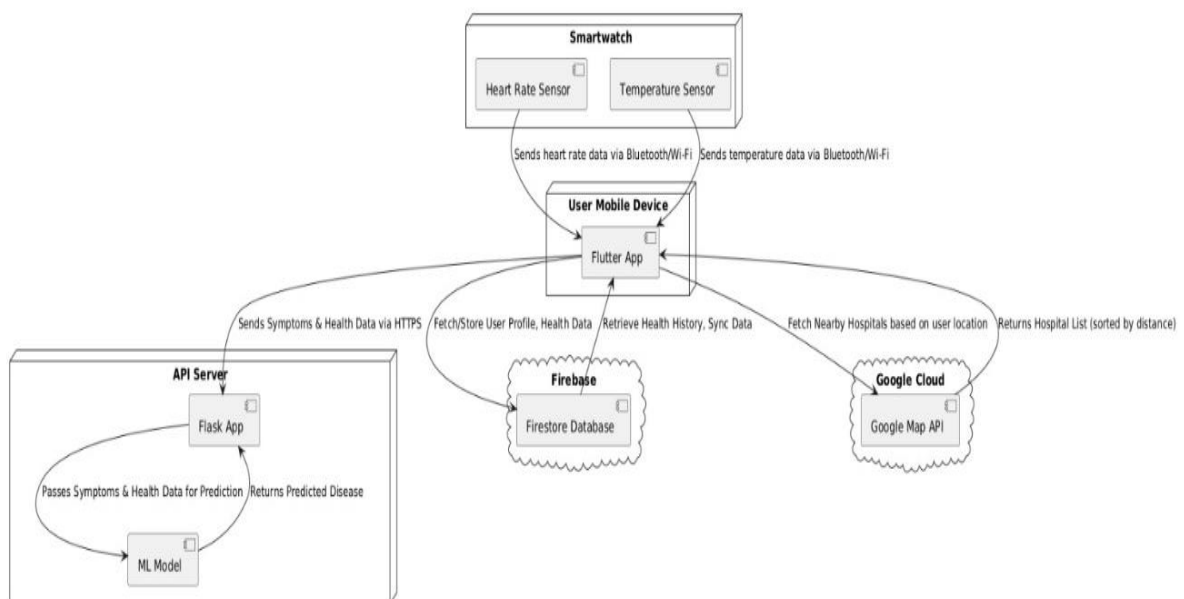


Figure 4.20 Deployment Diagram

Chapter 5

SOFTWARE DESIGN

CHAPTER 5:

Software Design

5.1 Introduction

This chapter outlines the development process of the ML-DDMSS system, the technologies and tools, the development environment, and the challenges faced during working. It gives an insight of the main modules and explains how they were developed and integrated into the system.

5.2 Product UI



Figure 5.1 Intro Screen

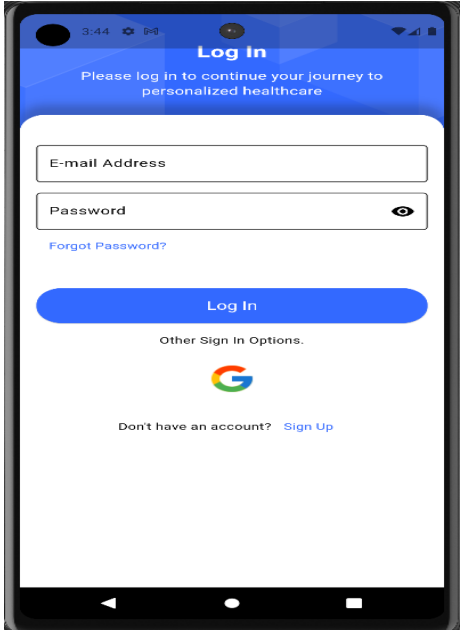


Figure 5.2 Login Screen



Figure 5.3 Home Screen

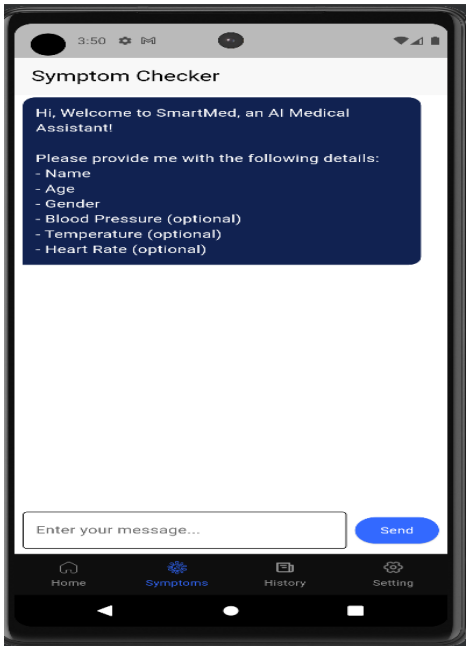


Figure 5.4 Enter Symptoms Screen



Figure 5.5 History Screen

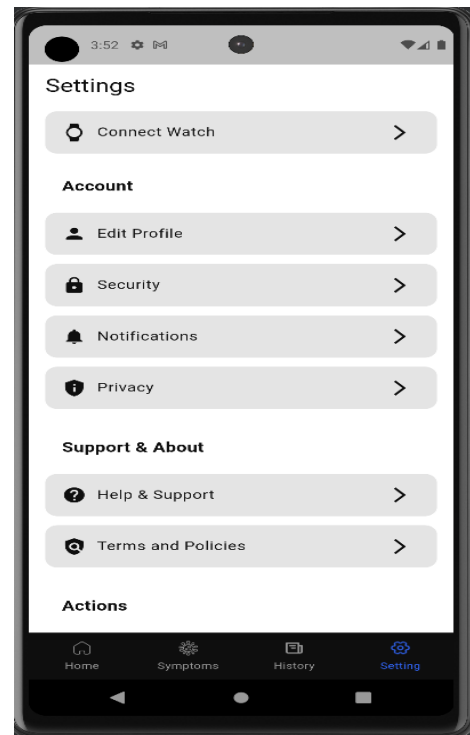


Figure 5.6 Setting Screen

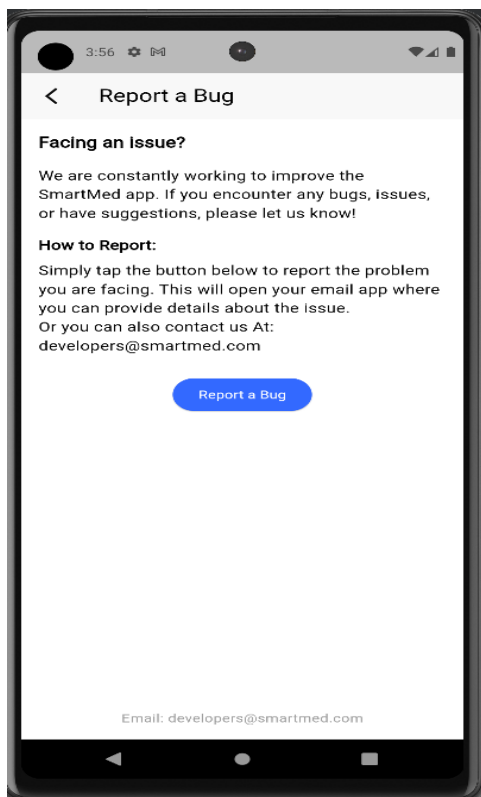


Figure 5.7 Help Screen

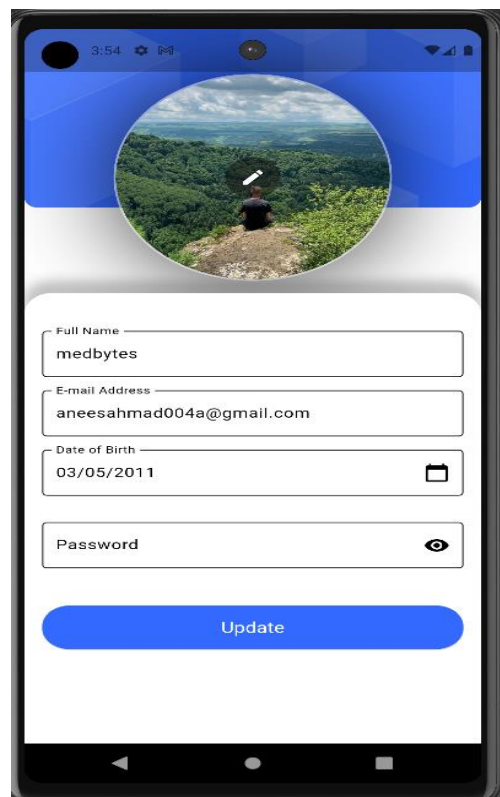


Figure 5.8 Profile Update Screen

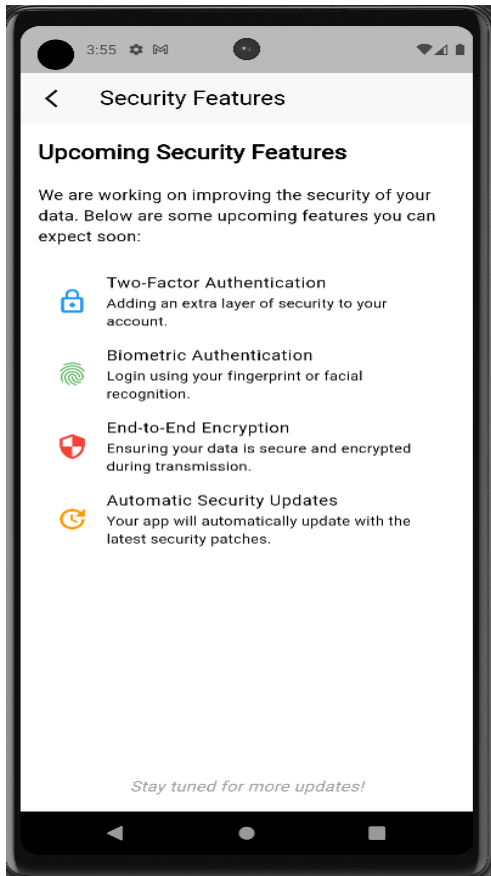


Figure 5.9 Security Screen

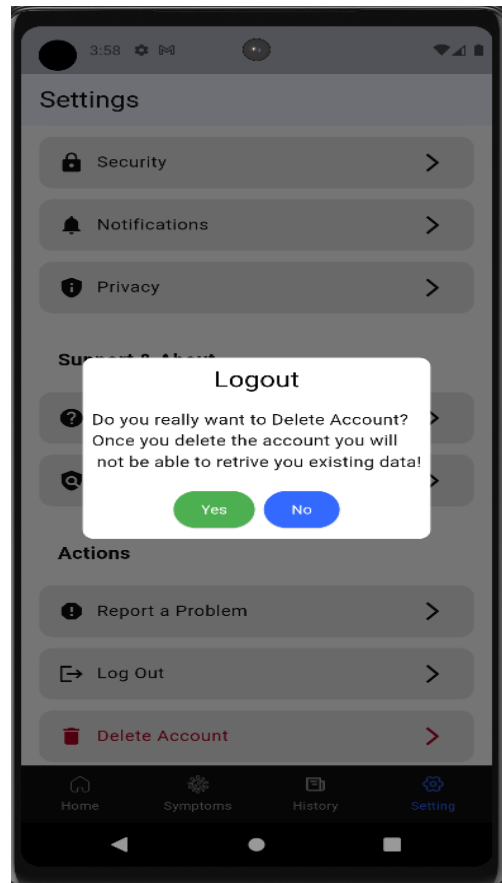


Figure 5.10 Logout Screen

Chapter 6

IMPLEMENTATION

CHAPTER 6:

Implementation

6.1 Introduction

This chapter outlines the development process of the ML-DDMSS system, the technologies and tools, the development environment, and the challenges faced during working. It gives an insight of the main modules and explains how they were developed and integrated into the system.

6.2 Development Environment

This section outlines the tools and technologies utilized during the creation of the ML-DDMSS system as described below:

- **Flutter:** Used for developing the mobile application, ensuring cross platform compatibility.
- **Python:** Utilized for developing the machine learning models used in disease prediction.
- **Firebase:** Used for real-time data collection and user authentication.
- **ChatGPT API:** Integrated for follow-up questions to collect the symptoms from user.
- **Google Maps API:** Integrated for providing hospital recommendations based on the user's location.

6.3 Dataset Used

We used data set DDXPlus that contains 1.3 million entries having initial evidences, pathonlogy, age, sex, etc.

We split our dataset in training, testing and validation with 0.8, 0.1 and 0.1 respectively. The two example field of dataset are given below:

Table 6. 1 Dataset entries

AGE	SEX	DIFFERENTIAL_DIAGNOSIS	PATHOLOGY	EVIDENCES	INITIAL_EVIDENCE
49	F	[['Bronchitis', 0.2023], ['GERD', 0.1792], ['Possible NSTEMI / STEMI', 0.1431], ['Unstable angina', 0.1345], ['Pericarditis', 0.0976], ['Anemia', 0.0971], ['Stable angina', 0.0749], ['Boerhaave', 0.0714]]	GERD	['E_53', 'E_54_@V_112', 'E_54@V_161', 'E_54@V_180', 'E_54@V_181', 'E_55@V_103', 'E_56@6', 'E_57@V_29', 'E_57@V_101', 'E_58@3', 'E_59@2', 'E_70', 'E_78', 'E_98', 'E_140', 'E_167', 'E_173', 'E_201']	E_201
70	F	[['URTI', 0.2126], ['Influenza', 0.1984], ['Bronchitis', 0.1492], ['HIV (initial infection)', 0.1120], ['Pneumonia', 0.1013], ['Tuberculosis', 0.0811], ['Chronic rhinosinusitis', 0.0684], ['Acute rhinosinusitis', 0.0442], ['Chagas', 0.0328]]	URTI	['E_41', 'E_50', 'E_53', 'E_54_@V_161', 'E_54@V_183', 'E_55@V_62', 'E_55@V_89', 'E_55@V_108', 'E_55@V_109', 'E_55@V_124', 'E_56@7']	E_201

6.3 Model Used for Training

We use machine learning model ‘Random Forest’. A random forest is a machine learning algorithm that combines multiple decision trees to produce a single result. It's a flexible, and easy-to-use algorithm.

Block Diagram of Model:

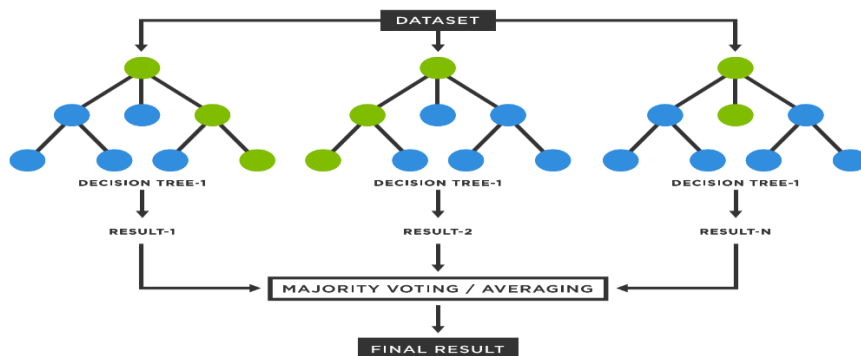


Figure 6.1 Random forest Block Diagram

6.4 Module Implementation

Each module of the ML-DDMSS system is described in detail, including its functionality, the technologies employed, and how it connects with other modules. Key modules include:

- **Symptom Input Module:** Allows users to input their symptoms for analysis.
- **Disease Diagnosis Module:** Analyses the symptoms and health data to generate disease predictions.
- **Real-time Data Collection Module:** Collects real-time health data from smartwatches and other IoT devices.
- **Hospital Recommendation Module:** Provides users with recommendations for nearby hospitals based on their predicted condition.

6.5 Challenges and Solutions

This section covers the challenges and solution addressed during the creation of the MLDDMSS system as shown below:

- Dealing with large dataset having 1.23 million entries.
- High Computational power required.
- Budget and Time management.
- Integrating real time health data with Model.
- Performance and Security.

Solutions:

To address the high computational power needs of the project, we leverage Google Colab with GPU support to ensure efficient model training while keeping costs minimal. Budget management is done due to PEC funding to our project. Integrate real-time health data with the model, we use google pixel smartwatch. Additionally, security is prioritized through encryption.

Chapter 7

TESTING AND VALIDATION

CHAPTER 7:

Testing and Validation

7.1 Introduction

The chapter on testing and validation outlines the methods for evaluating the ML-DDMSS system to ensure compliance with specifications. It provides an overview of the testing procedure, the test cases developed, and the results obtained.

7.2 Testing Methodology

This section details the testing techniques employed to validate the ML-DDMSS system, encompassing unit, integration, and system tests. Each test aims to verify that the system operates efficiently and meets the specified requirements.

7.3 Test Cases

Table 7.1 Signin using valid email password

Test Case	TC-01
Test Description	User sign-in functionality using email and password.
Input	Email: aneesahmad00423@gmail.com, Password: Smart@42672
Expected Result	User should be successfully signed in and redirected to the HomeScreen.
Actual Result	Login successful, HomeScreen displayed with recent readings.
Pass/Fail	Pass

Table 7.2 Signin using invalid email password

Test Case	TC-02
Test Description	Test sign-in with invalid email or password.
Input	Email: aneesahmad00423@gmail.com, Password: Smartmed@123
Expected Result	An error message should be shown stating 'Invalid credentials.'
Actual Result	Error message 'Invalid credentials' displayed.
Pass/Fail	Pass

Table 7.3 Signin with no email or password

Test Case	TC-03
------------------	-------

Test Description	Test sign-in with not email or password.
Input	Email: mujahidawan290@gmail.com, Password: Smartmed@42672
Expected Result	An error message should be shown stating 'Email is not verified.'
Actual Result	Error message 'Email is not verified' displayed.
Pass/Fail	Pass

Table 7.4 Signup using empty email or password

Test Case	TC-04
Test Description	Testing sign-up functionality if any of the field is empty.
Input	Name: “ ”, Email: realmrgeek @gmail.com, DOB: 17-03-2001 Gender: Male Password: SmartMed42672##
Expected Result	An error message should be shown stating ‘Can not sign-up with empty name.'
Actual Result	An error message is displayed.
Pass/Fail	Pass

Table 7.5 Signup using short password

Test Case	TC-05
Test Description	Sign-up functionality if password is short and weak.
Input	Name: medbytes, Email: realmrgeek @gmail.com, DOB: 17-03-2001 Gender: Male Password: smartmed
Expected Result	An error message should be shown stating ‘Password must be at least 6 chars long. With at least one capital letter, one small letter, one digit and one special char'
Actual Result	An error message is displayed.
Pass/Fail	Pass

Table 7.6 Signup using email already exist

Test Case	TC-06
Test Description	Sign-up functionality if email already exist.

Input	Name: medbytes, Email:aneesahmad004a@gmail.com, DOB: 17-03-2001 Gender: Male Password: smartmed
Expected Result	An error message should be shown stating 'email already exist.'
Actual Result	An error message is displayed.
Pass/Fail	Pass

Table 7.7 Signup with valid details

Test Case	TC-07
Test Description	sign-up functionality for a new user.
Input	Name: Smartmed Dev, Email: realmrgeek @gmail.com, DOB: 17-03-2001 Gender: Male Password: SmartMed42672##
Expected Result	The user should be successfully signed up and redirected to the Sign In Screen with message verification email sent.
Actual Result	Sign up successful, Sign In Screen displayed with message.
Pass/Fail	Pass

Table 7.8 PDF generation

Test Case	TC-08
Test Description	Test the functionality to generate a PDF of the health reading history.
Input	User clicks on 'Generate PDF' button from the history screen.
Expected Result	A PDF with recent health readings should be generated and available for download.
Actual Result	PDF generated successfully and available for download.
Pass/Fail	Pass

Table 7.9 Reset password with valid email

Test Case	TC-09
Test Description	Reset password if user forgot
Input	User input the email and click send button.
Expected Result	Sign In screen should be displayed with message "reset email sent".
Actual Result	Received and password reset.

Pass/Fail	Pass
------------------	------

Table 7.10 Reset password using invalid email

Test Case	TC-10
Test Description	Reset password with invalid email.
Input	User input the invalid email and click send button.
Expected Result	Error message should be displayed stating “input email is not valid”
Actual Result	Received and password reset.
Pass/Fail	Pass

Table 7.11 Signin with google

Test Case	TC-11
Test Description	Test the Google sign-in functionality.
Input	Google account credentials
Expected Result	The user should be successfully signed in with their Google account and redirected to the HomeScreen.
Actual Result	Google login successful, HomeScreen displayed.
Pass/Fail	Pass

Table 7.12 When no record on database

Test Case	TC-12
Test Description	Handle the case if no readings available in database
Input	Access the Home Screen after sign-in.
Expected Result	Show a text with No reading recorded
Actual Result	Text displayed.
Pass/Fail	Pass

Table 7.13 Recent smartwatch reading

Test Case	TC-13
Test Description	Recent health readings are displayed in real time (change in firebase displayed without loading).
Input	Access the Home Screen after sign-in.
Expected Result	Recent health readings (e.g., heart rate, temperature) and last reading time should be changed in real-time.

Actual Result	Health readings and graph are changing in real-time.
Pass/Fail	Pass

Table 7.14 Hospitals recommendation

Test Case	TC-14
Test Description	Test the functionality of showing recommended hospitals based on location.
Input	User clicks on 'Show Hospitals' after disease prediction.
Expected Result	A list of nearby hospitals should be displayed with distance and rating.
Actual Result	Nearby hospitals displayed with details (distance, rating).
Pass/Fail	Pass

Table 7.15 Connecting smartwatch

Test Case	TC-15
Test Description	Test the functionality of connecting a smartwatch.
Input	User tries to connect a smartwatch by signing in email and password.
Expected Result	The smartwatch should be successfully connected, and redirect to home screen.
Actual Result	Smartwatch connected successfully,
Pass/Fail	Pass

Table 7.16 Symptoms input

Test Case	TC-16
Test Description	Test the symptom input functionality.
Input	symptoms: "fever", "chills", "sweats", "headache", "fatigue",
Expected Result	The symptoms should be accepted and processed by the system.
Actual Result	Symptoms input accepted successfully.
Pass/Fail	Pass

Table 7.17 Model prediction

Test Case	TC-17
Test Description	Test the disease prediction functionality using ML model.

Input	symptoms: "fever", "chills", "sweats", "headache", "fatigue",
Expected Result	The ML model should predict a possible disease (e.g., Anemia) and display it on the screen.
Actual Result	ML model predicted 'Anemia' and displayed the result.
Pass/Fail	Pass

Table 7.18 App initial lunch

Test Case ID	TC-18
Test Description	Test the initial launch of the app and the display of the Intro Screen
Input	Launch the app for the first time
Expected Result	The Intro Screen should be displayed with some information about application.
Actual Result	Intro Screen displayed
Pass/Fail:	Pass

Table 7.19 Lunch for existing user

Test Case	TC-19
Test Description	Test the launch of the app if user already logged in.
Input	Launch the app.
Expected Result	The home should be displayed with user profile and other data.
Actual Result	Home displayed successfully.
Pass/Fail	Pass

Table 7.20 Update profile

Test Case	TC-20
Test Description	Update Profile
Input	Name: medbytes2, DOB: 17-03-2002 Gender: Male Password: Smartmed@123
Expected Result	Profile should be updated and with success message.
Actual Result	Profile updated success
Pass/Fail	Pass

Table 7.21 Logout

Test Case	TC-21
Test Description	Test the logout functionality.
Input	User clicks on 'Logout' in the settings menu.

Expected Result	The user should be logged out and returned to the Sign in screen.
Actual Result	User logged out successfully and returned to the Sign In screen.
Pass/Fail	Pass

7.4 Testing Validation

The test results are shown, demonstrating that the ML-DDMSS system meets the requirements and works as expected. Any issues encountered during testing are addressed. Firstly, system is validated by the development team using a manual testing to check against requirement specification document.

CONCLUSION AND FINDINGS

CHAPTER 8:

Conclusion and findings

8.1 Conclusion and Findings

The last chapter summarizes the ML-DDMSS project's findings, emphasizing the system's ability to increase healthcare accessibility and efficiency through the integration of machine learning and IoT technology. It addresses the system's potential to improve healthcare delivery, particularly in distant and disadvantaged areas, by providing early and accurate disease diagnosis and medical treatment recommendations.

8.1.1 Findings

Here are the findings of our project that represents the accuracy, precision, F1-score, heat map, confusion matrix and histogram given below:

Table 8.1 Performance matrix

PERFORMANCE MATRIX	VALUE
Accuracy	99%
F1-score	99%
Precision	99%

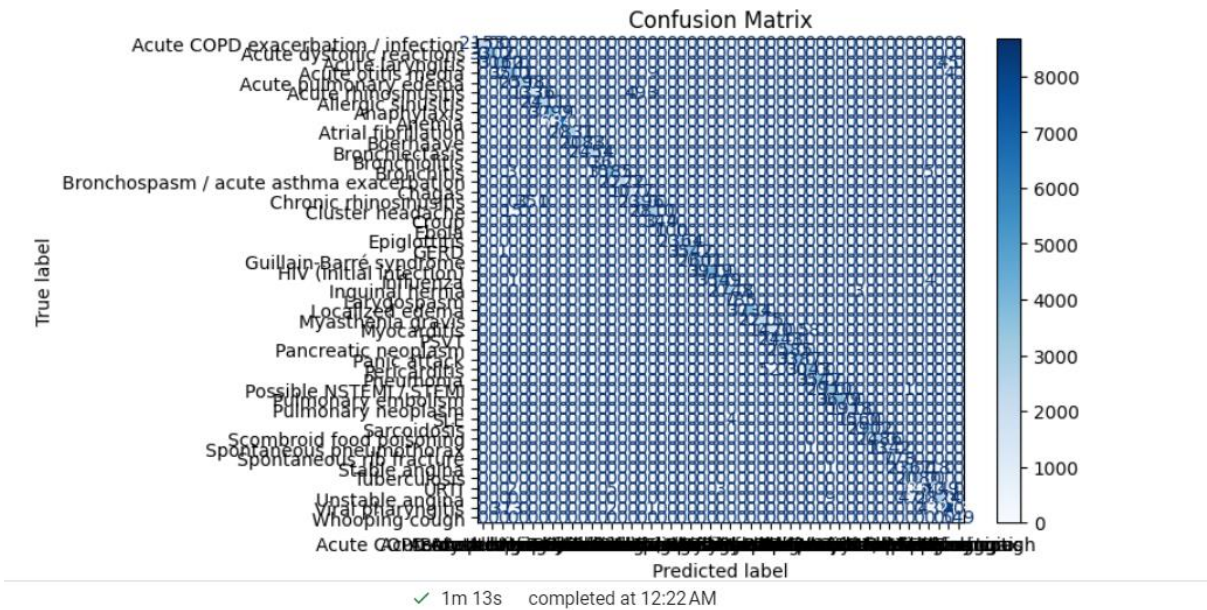


Figure 8.2 Confusion matrix

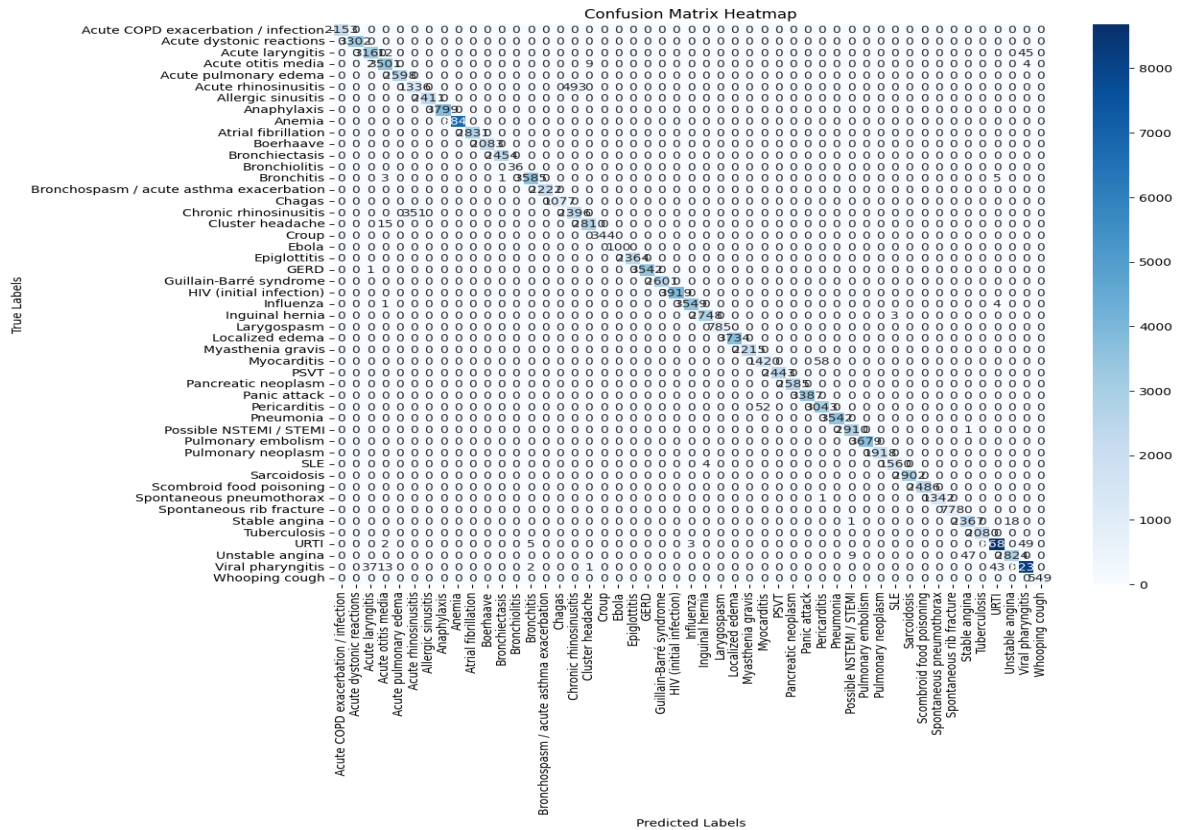


Figure 8.3 Heat map

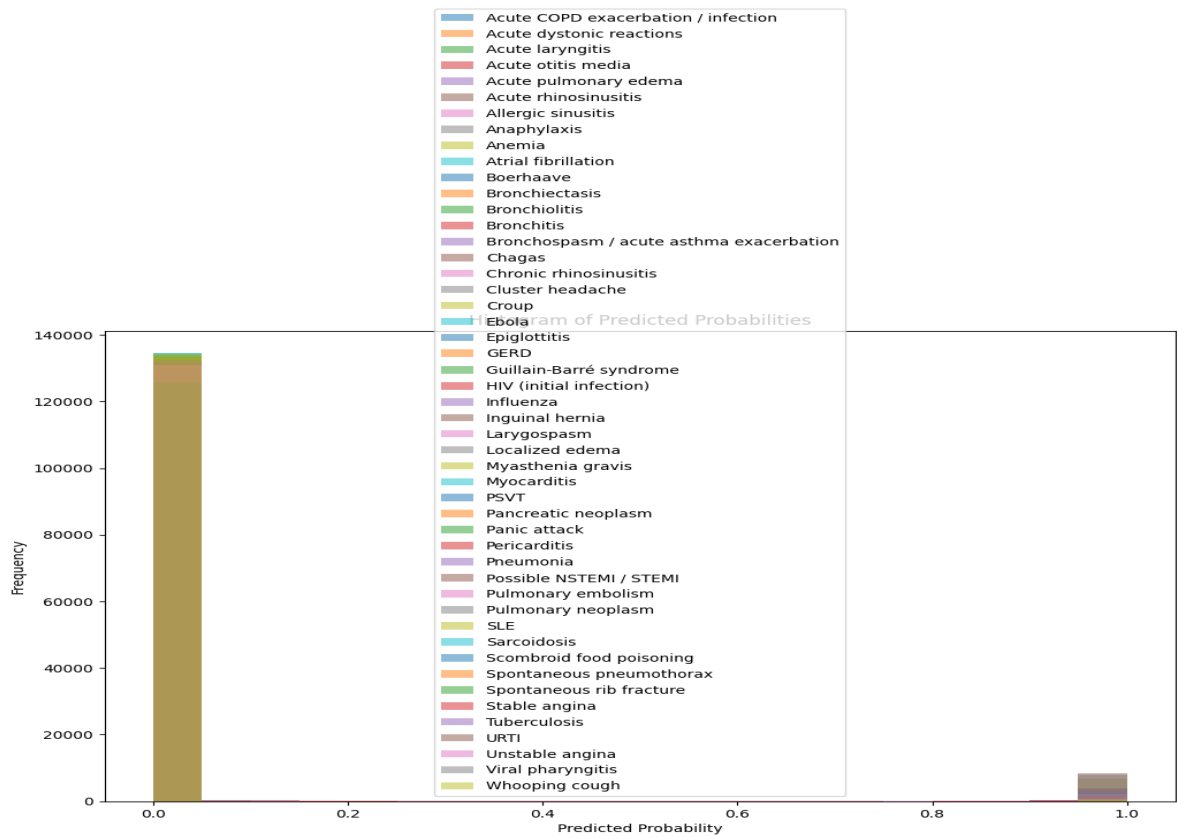
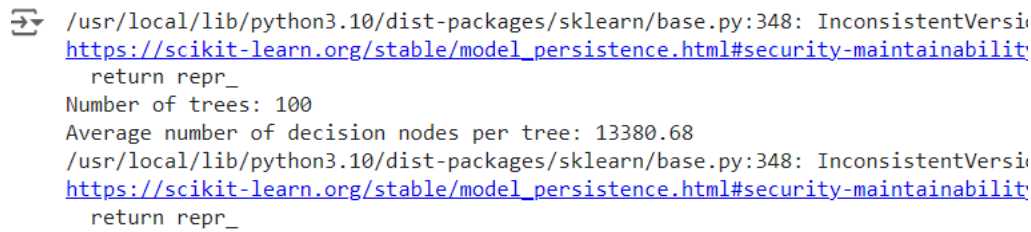
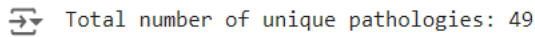


Figure 8.4 Histogram



```
➦ /usr/local/lib/python3.10/dist-packages/sklearn/base.py:348: InconsistentVersionWarning:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability
return repr_
Number of trees: 100
Average number of decision nodes per tree: 13380.68
/usr/local/lib/python3.10/dist-packages/sklearn/base.py:348: InconsistentVersionWarning:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability
return repr_
```

Figure 8.5 No of trees and node



```
➦ Total number of unique pathologies: 49
```

Figure 8.6 Total no of pathologies

8.2 Limitations and Future Work

This section highlights potential future development areas, such as expanding the system to support other health data sources, incorporating more advanced machine learning models, and exploring new applications for the system in other areas of healthcare. The system may be scaled to suit larger populations and more sophisticated medical challenges.

8.2.1 Limitations in our system:

- Not suitable for critical patients.
- Lack of doctor appointment.
- Lack of personalized predictions.
- Lack of voice chatting.

8.2.2 Future work:

- Doctor appointment.
- Personalized predictions.
- Voice chatting for better understanding of patient feelings.

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